

PAPER_1491_NI62_PEAK_BINDING

Daniel T. Murphy

PAPER_1491 — UQFF: Ni-62 Peak Nucleon Binding (extends PAPER_872) (Bucket G)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket G — derived from F:\Aetheric Propulsion source material

Date: June 16, 2026

Location: 41.0997 N, 80.6495 W (Youngstown, OH, USA)

Status: CLOSED — $Z = D_{\text{crit}} + 2 = 28$, $N = D_{\text{crit}} + 2 \cdot D_{\text{phys}} = 34$, $A = A_5 + 2 = 62$ — all EXACT

Observation

Nickel-62 has the highest binding energy per nucleon in nature ($BE/A = 8.7945$ MeV, surpassing Fe-56 at 8.790 MeV).

UQFF Closed Identity

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$Z(\text{Ni-62}) = D_{\text{crit}} + 2 = 28$ EXACT; $N(\text{Ni-62}) = D_{\text{crit}} + 2 \cdot D_{\text{phys}} = 34$ EXACT;

$A(\text{Ni-62}) = A_5 + 2 = 62$ EXACT.

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Physical Interpretation

Ni-62 extends the proto-element chain established in PAPER_872 (proto-Fe $Z = D_{\text{crit}} = 26$ EXACT; proto-Si $Z = SO_5 + D_{\text{phys}} = 14$ EXACT). The +2 offset from D_{crit} reflects the additional pairing energy above the proto-Fe shell closure. All three nuclear quantum numbers (Z , N , A) reduce to canonical integer-primitive identities.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical UQFF integer primitives $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$. Zero free parameters.

Reference

- Source: F:\Aetheric Propulsion (Daniel's reactor + experimental docs, 2025-2026)
- Related: PAPER_646 (Universal Inertial Operator, Caduceus, DPM mechanism), PAPER_872 (proto-element nuclear identity), PAPER_087 (BH thermodynamics), PAPER_1156 (cosmology suite), PAPER_1255 (muonic hydrogen radius).

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